StudyQuest AI: A Game-Based Study Planner

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***Abstract*—The increasing academic workload and the lack of effective study planning make it difficult for students to manage their time, understand complex subjects, and stay motivated during exam preparation. This paper presents StudyQuest AI, an interactive web application that combines study planning with gamification to make learning more engaging and organized. The system provides a dashboard that tracks experience points (XP), completed quests, and study streaks. Users generate a study quest route by selecting a subject, available study time, exam date, and planning strategy, and the application divides the learning process into structured activities: concept learning, practice sessions, quizzes, and a boss battle. Game-based elements such as levels, rewards, achievement badges, progress tracking, countdown timers, and interactive quizzes are incorporated to encourage consistent study habits. In the boss battle, a correct answer defeats the opponent and rewards the user with additional XP, while an incorrect answer reduces the opponent's health points. The application is developed using HTML, CSS, and JavaScript with a responsive user interface and rule-based interactions; the current implementation uses predefined quest data and fixed quest-generation logic rather than a trained machine learning model or an external AI service. Functional testing confirms that the modules work together as intended. StudyQuest AI demonstrates how gamification and structured study activities can support student motivation, task completion, and time management, and future work includes personalized recommendations, adaptive scheduling, performance analytics, and machine learning integration.**

***Index Terms*—educational technology, game-based learning, gamification, rule-based systems, student motivation, study planning, web application**

I. Introduction

Students often face difficulties in managing their academic workload, organizing study schedules, and maintaining motivation throughout the learning process. Traditional study-planning methods depend mainly on manually created timetables and task lists, which may not provide enough engagement or encouragement to sustain consistent study habits. Gamification, the use of game-design elements in non-game contexts [1], offers a way to make such routine activities more engaging, while self-determination theory identifies autonomy, competence, and relatedness as key factors that influence motivation [2].

This paper presents StudyQuest AI, a game-based study planner that combines study planning with gamification to create a more interactive learning experience. The system is a web-based application that allows students to organize their study activities as quests and to track their progress through a dashboard. A student selects a subject, specifies the available study time, enters an exam date, and chooses a planning strategy to generate a structured quest route. The route consists of learning concepts, practicing problems, attempting quizzes, and completing a final boss battle, dividing the learning process into manageable tasks that the student completes step by step.

StudyQuest AI incorporates game-based features such as experience points (XP), levels, achievement badges, study streaks, progress indicators, countdown timers, and interactive challenges. Completing quests increases the student's XP and completed-quest count, while the boss battle provides an additional question-based challenge with rewards. These features aim to encourage regular participation, improve task completion, and make study sessions more enjoyable.

The application is implemented using HTML, CSS, and JavaScript and provides dashboard navigation, quest management, timer controls, quiz interaction, and reward tracking. The current version uses predefined quest information and rule-based functionality to generate study routes and manage game progress; it does not implement a trained machine learning model or connect to an external AI service. The work therefore demonstrates the practical application of gamification and structured task planning in an educational setting.

The remainder of this paper is organized as follows. Section II reviews related work. Section III describes the proposed system and its architecture. Section IV details the system modules. Section V presents the implementation and results, and Section VI concludes the paper and outlines future work.

II. Related Work

This section reviews nine studies covering gamification, student motivation, effective learning techniques, and artificial intelligence foundations. Table I summarizes the methodology of each study, its principal limitation, and how the proposed system applies the relevant concepts.

Deterding *et al.* [1] introduced the concept of gamification by applying game-design elements in non-game contexts. Their work provides a conceptual foundation but does not present a complete study-planning application. Deci and Ryan [2] examined self-determination theory, focusing on autonomy, competence, and relatedness as factors influencing motivation; the theory, however, does not provide a specific software implementation for academic planning. Kapp [3] explained the application of game-based learning, game mechanics, and instructional design in education and training, offering broad instructional guidance rather than a personalized study-planning system.

Hamari *et al.* [4] reviewed empirical studies on the effects of gamification and reported that outcomes vary depending on the context and implementation. Sailer *et al.* [5] conducted an experimental study of how specific game-design elements influence psychological need satisfaction and motivation, examining selected elements in an experimental setting rather than implementing a complete study planner. Dichev and Dicheva [6] critically reviewed gamification in education and highlighted uncertainty regarding its long-term educational benefits.

Dunlosky *et al.* [7] reviewed learning techniques such as practice testing and distributed practice to evaluate their effectiveness for student learning, focusing on the techniques themselves rather than on an integrated, gamified study-planning application. Werbach and Hunter [8] explained how game elements, mechanics, and dynamics can be used to design gamified systems, providing a general framework that does not specifically address automated academic quest generation. Russell and Norvig [9] presented foundational artificial intelligence concepts, including intelligent agents, problem-solving, search strategies, and decision-making, which constitute general AI theory rather than a ready-made educational gamification platform.

TABLE I
SUMMARY OF RELATED WORK

Work	Methodology and Limitation	Application in StudyQuest AI
Deterding et al. [1]	Defined gamification as the use of game-design elements in non-game contexts; conceptual, with no study-planning application.	Applies gamification through quests, XP, levels, badges, and progress tracking.
Deci and Ryan [2]	Self-determination theory: autonomy, competence, and relatedness drive motivation; no software implementation for academic planning.	Selectable study preferences, achievable quests, and progress indicators to support motivation.
Kapp [3]	Game-based learning, game mechanics, and instructional design; broad guidance rather than a personalized planner.	Combines structured learning tasks with interactive challenges and rewards in one web application.
Hamari et al. [4]	Review of empirical gamification studies; outcomes vary with context and implementation.	Integrates quest completion, XP, streaks, and achievements to support study engagement.
Sailer et al. [5]	Experiment on specific game-design elements and need satisfaction; does not implement a complete study planner.	Incorporates progress feedback, achievement rewards, and challenge-based activities in the study workflow.
Dichev and Dicheva [6]	Critical review of gamification in education; long-term educational benefits remain uncertain.	Organizes study tasks and makes progress visible through an interactive dashboard.
Dunlosky et al. [7]	Evaluated learning techniques such as practice testing and distributed practice; not an integrated gamified planner.	Quiz-based challenges and structured study sessions that support active practice.
Werbach and Hunter [8]	Framework of game elements, mechanics, and dynamics; does not address automated academic quest generation.	Uses quests, XP, badges, and boss battles to structure the learning journey.
Russell and Norvig [9]	Foundational AI concepts: intelligent agents, search, and decision-making; general theory, not an educational platform.	Foundation for future intelligent recommendations and adaptive planning.

A common theme across these studies is that theoretical frameworks explain motivation and learning, while many gamification approaches focus on individual game elements rather than integrating them into a structured study-planning application. Research on self-determination theory highlights the importance of motivation and competence, and studies on effective learning techniques emphasize the value of practice testing and organized learning activities. These concepts alone, however, do not provide students with a single interactive platform for organizing study tasks and tracking their progress.

StudyQuest AI addresses this gap by combining study planning and gamification within one web application. It provides a quest-based dashboard, subject selection, study-time and exam-date inputs, structured learning activities, countdown timers, quiz challenges, XP, and achievement badges. Unlike a conventional timetable or task list, the system presents study activities as progressive challenges that students can complete and track.

III. PROPOSED SYSTEM

A. Overview

StudyQuest AI provides an interactive study-planning experience by combining structured learning activities with game-based elements. Instead of relying entirely on traditional timetables and task lists, the system organizes academic activities into quests that students complete while tracking their progress and achievements.

The dashboard shows the student's XP, completed quests, study streaks, and active study activities. From the Quest Map, the student selects a subject, available study time, exam date, and preferred planning strategy. Based on these inputs, the system generates a predefined quest route consisting of concept learning, practice activities, an AI quiz, and a boss battle. The Quest Detail view shows the description, estimated duration, and XP reward of a selected quest, together with a countdown timer offering start, pause, and reset controls. The student must complete three learning objectives (reading the concept, practicing examples, and completing the quiz) before the Complete Quest button becomes available; on completion, the application updates the XP and completed-quest count and displays a confirmation message.

The Missions view introduces a quiz challenge in which the student answers questions to defeat a virtual opponent. A correct answer defeats the boss and awards additional XP, while an incorrect answer reduces the opponent's health points. The Rewards view displays achievement badges such as Quest Master, Streak Warrior, Quiz Slayer, and Boss Hunter, encouraging students to work toward learning milestones.

B. System Architecture

The architecture of StudyQuest AI is shown in Fig. 1. It is organized into four layers. The frontend layer (HTML/CSS) contains the user-interface components, the study-planner inputs, the interactive pages, and the progress-tracking display with which the user interacts through a web browser. The application logic layer (JavaScript) comprises the navigation,

quest-generation, timer, quiz, and progress-and-reward modules, which process user input and events. The data layer holds the predefined subject, quest, and quiz data, including topics, learning objectives, question sets, correct answers, and XP rewards. The presentation layer renders the Dashboard, Quest Map, Quest Detail, and Rewards views.

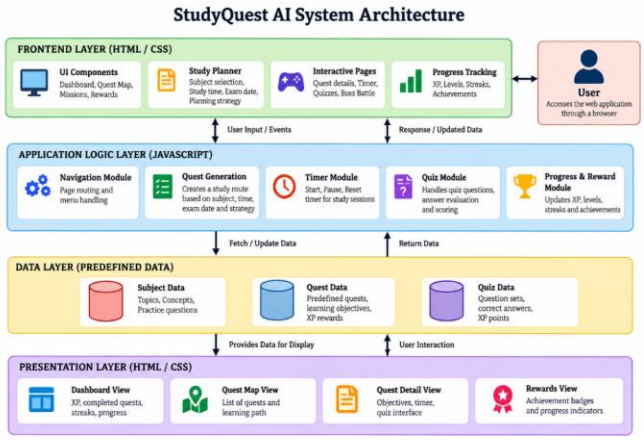


Fig. 1. Architecture of StudyQuest AI.

When the student enters study preferences on the Quest Map, the JavaScript logic processes the inputs and generates the predefined quest route. Quest details, the study timer, task completion, quiz interactions, and reward calculations are all managed by this logic, which updates the XP and completed-quest count and returns feedback to the interface. All components work together through the HTML, CSS, and JavaScript implementation. The current version uses predefined quest data and does not include a backend server or database for persistent storage.

IV. Module Description

StudyQuest AI is divided into five modules, each performing a specific function: dashboard management, quest generation, study-session tracking, interactive quiz challenges, and reward management.

A. Dashboard and Navigation

This module provides the main interface through which students access the features of StudyQuest AI. The dashboard displays the student's XP, completed quests, current study streak, and active quests. The navigation system allows users to move between the Dashboard, Quest Map, Missions, and Rewards pages, giving a centralized view of study progress.

B. Study Planning and Quest Generation

This module enables students to generate a structured study route by selecting a subject, available study time, exam date, and planning strategy. Based on the selected inputs, the application generates a predefined sequence of activities: Concept Quest, Practice Battle, AI Quiz, and Boss Battle. Each activity represents a stage in the learning process. The current implementation uses predefined quest-generation logic rather than a machine learning model or an adaptive AI algorithm.

C. Quest Management and Study Timer

This module manages individual quests and helps students monitor their study sessions. Each quest displays its title, description, estimated duration, and XP reward, and a countdown timer supports start, pause, and reset operations. The module includes three learning objectives: reading the concept, practicing examples, and completing the quiz. The Complete Quest button is enabled only after all three objectives have been selected. Once the quest is completed, the application updates the XP total and completed-quest count and displays a confirmation message.

D. Missions and Boss Battle

This module introduces a question-based challenge. The Missions page presents a Python function question with multiple answer choices and a virtual opponent called the Code Goblin, which begins with 100 health points. A correct answer defeats the opponent and awards additional XP, provided the boss has not already been defeated. An incorrect answer reduces the opponent's health by 20 points and displays feedback. The module thus combines quiz interaction with game-based challenges to encourage active participation.

E. Rewards and Progress Tracking

This module displays the student's progress and achievement badges. The application tracks XP, completed quests, and progress toward the next XP milestone, and provides achievement cards such as Quest Master, Streak Warrior, Quiz Slayer, and Boss Hunter, each representing a specific milestone. Progress values are managed in JavaScript during application use; the current version does not include a database for permanent progress storage.

V. Implementation and Results

A. Experimental Setup

StudyQuest AI is implemented using HTML [10], CSS [11], and JavaScript [12]. HTML defines the structure of the application pages, including the Dashboard, Quest Map, Missions, and Rewards. CSS provides the visual design, layout, colors, and responsive styling. JavaScript manages page navigation, predefined quest data, quest-route generation, countdown-timer operations, learning-objective validation, quiz interactions, XP calculations, and progress updates. The implementation uses predefined quest information and rule-based JavaScript logic; it does not use a trained machine learning model, an external AI service, or a backend database.

The application was tested through its interactive features to verify that students can navigate between pages, generate a quest route by entering study preferences, open individual quests, operate the study timer, complete the required learning objectives, answer the Python function quiz in the boss battle, and view achievement badges, with the dashboard reflecting updates to XP and completed quests.

B. Results

The dashboard (Fig. 2) displays the student's current XP, completed quests, study streak, and active study activities,

providing a centralized view of progress and navigation to the Quest Map, Missions, and Rewards modules.

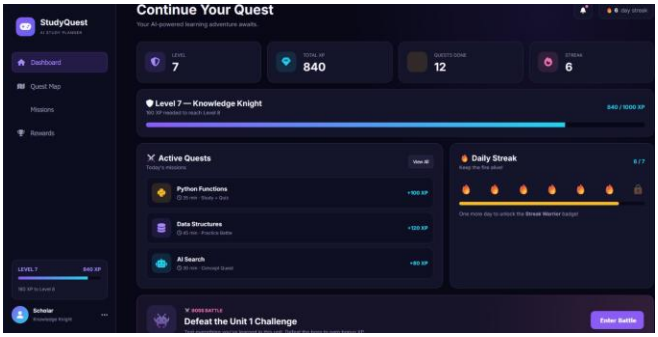


Fig. 2. StudyQuest AI dashboard.

The Quest Map (Fig. 3) allows students to select a subject, enter their available study time and exam date, and choose a planning strategy. The quest-generation module successfully creates the predefined study route of concept learning, practice activities, an AI quiz, and a boss battle from the selected inputs.

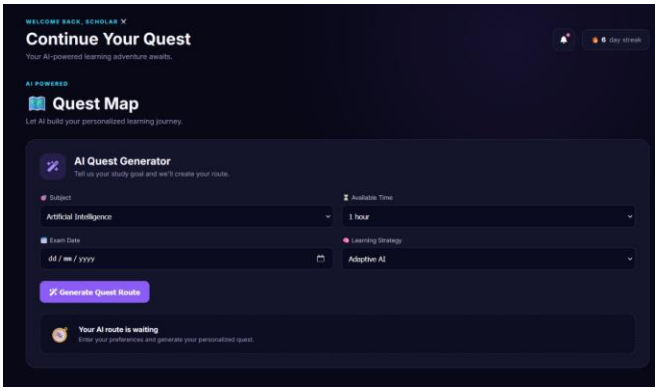


Fig. 3. Quest Map and study route generation.

The Quest Detail page (Fig. 4) displays the selected quest's title, description, estimated duration, and XP reward. The countdown timer supports start, pause, and reset operations, and the module verifies that all three learning objectives are selected before enabling the Complete Quest button. After a quest is completed, the application updates the XP and completed-quest count and displays a confirmation message.

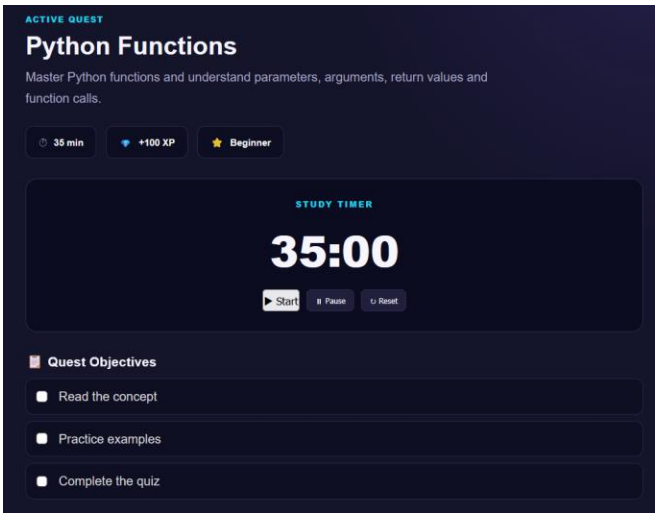


Fig. 4. Quest details and study timer.

The Missions module (Fig. 5) presents an interactive Python quiz and the Code Goblin opponent. A correct answer defeats the opponent and awards additional XP if it has not already been defeated, while an incorrect answer reduces its health points, demonstrating the integration of quiz-based learning and game mechanics.

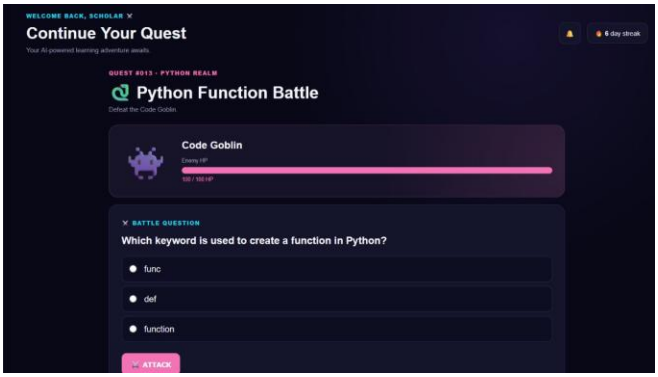


Fig. 5. Missions and boss battle.

The Rewards module (Fig. 6) displays the achievement badges Quest Master, Streak Warrior, Quiz Slayer, and Boss Hunter, while the dashboard presents updated progress information whenever quests are completed.

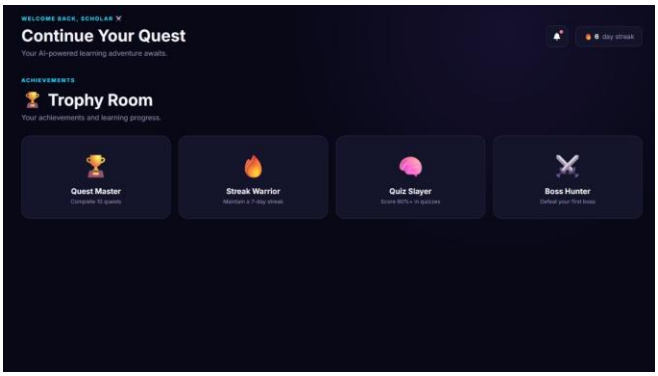


Fig. 6. Rewards and progress tracking.

C. Discussion

The results show that the individual components of StudyQuest AI work together as intended and that web technologies combined with rule-based interactions can produce a functional gamified study planner. The application gives students a structured way to organize study activities, monitor quest completion, and engage with learning challenges. However, formal measurements of improvements in academic performance, long-term motivation, or study efficiency have not yet been established, and the quest route is generated from predefined data rather than adapted to the individual student. Future testing with students and the collection of usage data are required to evaluate the educational effectiveness of the system.

VI. Conclusion and Future Work

This paper presented StudyQuest AI, a web-based game-based study planner designed to make study planning more interactive and engaging through gamification. By combining structured study quests, countdown timers, interactive quizzes, XP, achievement badges, and progress tracking, the system provides students with an organized approach to managing their learning activities. The application uses HTML, CSS, and JavaScript, with predefined quest data and rule-based interactions. Testing of the implemented features demonstrated that students can generate study quest routes from their selected preferences, complete learning objectives, manage study sessions with the timer, and participate in quiz-based boss battles, with XP and completed-quest counts updated and visual feedback provided through the dashboard and rewards interface. The current implementation does not include a trained machine learning model or an adaptive AI engine, and its effectiveness in improving academic performance has not yet been formally evaluated.

As future work, StudyQuest AI can be enhanced with personalized study recommendations based on student performance, adaptive quest generation according to topic difficulty, and automated scheduling based on exam dates and available study hours. Database integration would enable persistent storage of student progress, completed quests, and achievements across sessions. Additional features could include a larger question bank, subject-specific quizzes, detailed performance analytics, and reminders for upcoming study sessions. Further evaluation with students would help measure the application's impact on study consistency, motivation, and time management.

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